

The generalized Gaussian iterated map

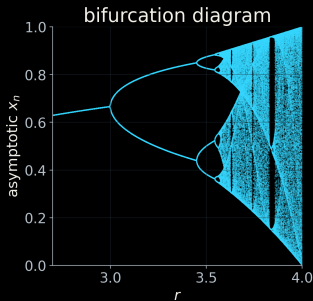
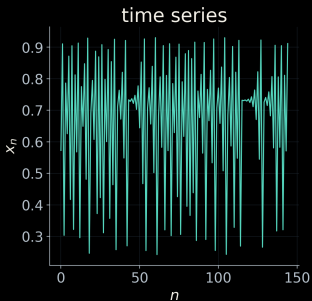
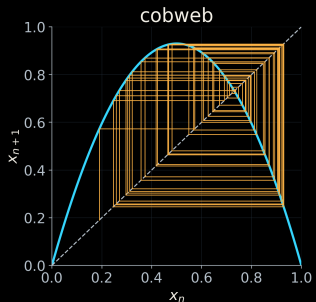
explicit geometry, bifurcations, and statistical structure



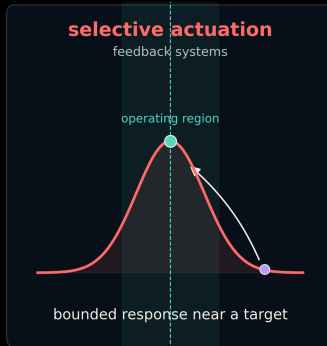
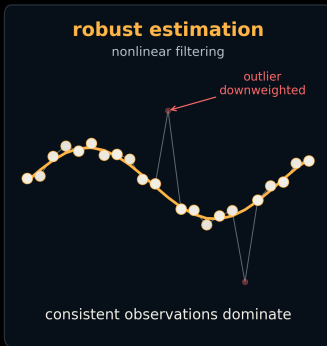
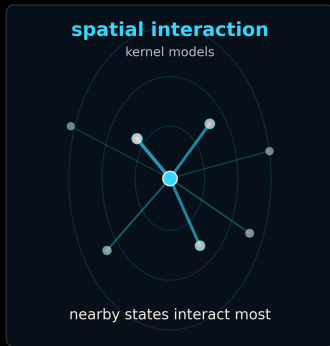
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The logistic map sets the benchmark

$$x_{n+1} = rx_n(1 - x_n), \quad x_n \in [0, 1].$$

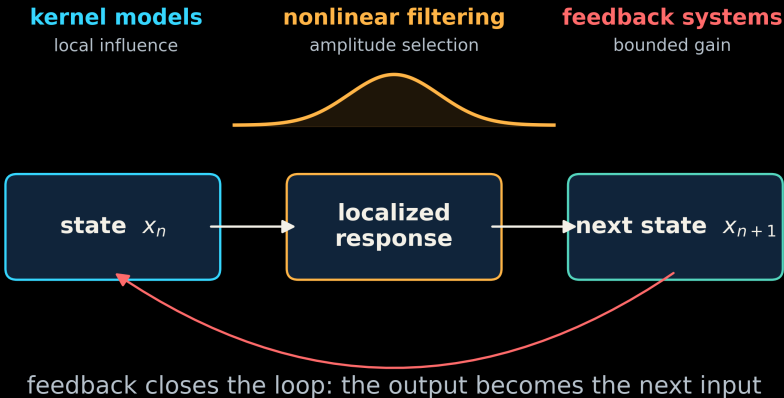


Applications call for localized feedback



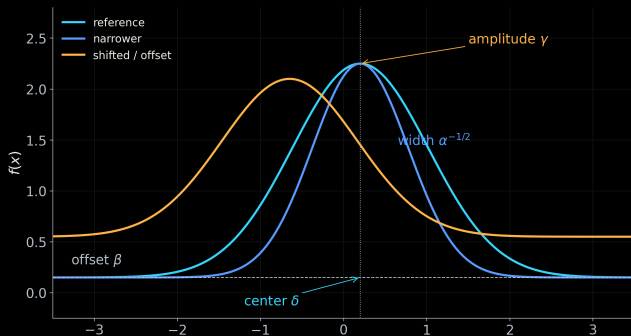
strong near a target • rapidly decaying influence away from it

Localized feedback recurs

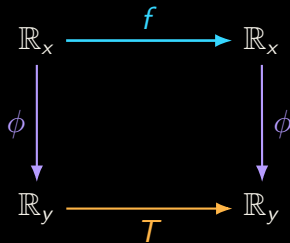


Four parameters shape the response

$$x_{n+1} = \gamma e^{-\alpha(x_n - \delta)^2} + \beta.$$



Four parameters collapse to two



$$y = \phi(x) := \sqrt{\alpha}(x - \delta),$$

$$A = \sqrt{\alpha}(\beta - \delta), \quad B = \sqrt{\alpha}\gamma.$$

$$\phi \circ f = T \circ \phi,$$

$$T(y) = A + Be^{-y^2}.$$

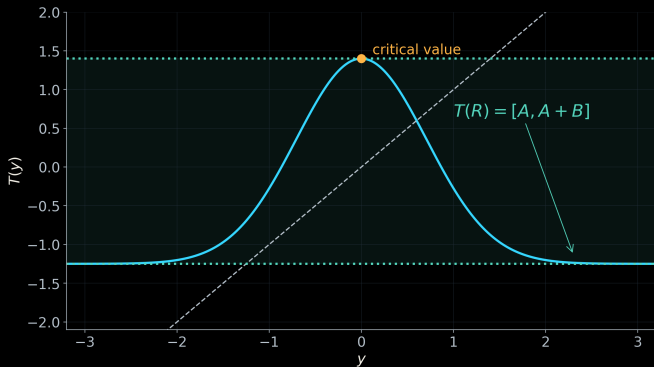
Affine conjugacy in one line

With $y = \phi(x) = \sqrt{\alpha}(x - \delta)$,

$$\begin{aligned}(\phi \circ f)(x) &= \sqrt{\alpha} \left(\gamma e^{-\alpha(x-\delta)^2} + \beta - \delta \right) \\ &= \underbrace{\sqrt{\alpha}(\beta - \delta)}_A + \underbrace{\sqrt{\alpha}\gamma}_{B} e^{-y^2} = (T \circ \phi)(x).\end{aligned}$$

Hence $T^n = \phi \circ f^n \circ \phi^{-1}$; periods and multipliers correspond.

The normalized map is a bounded fold



► $T(\mathbb{R}) = (A, A + B]$

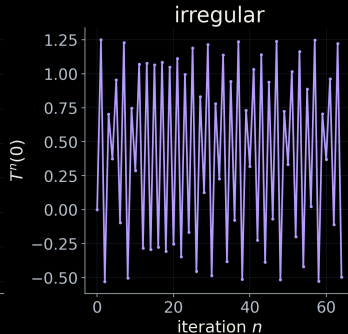
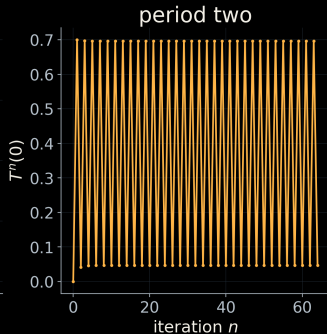
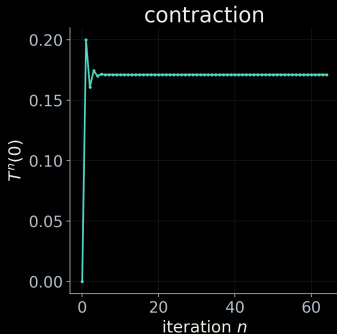
► $I = [A, A + B]$ is invariant

► unique critical point $c = 0$

► $T''(0) = -2B \neq 0$

$$T(y) = A + B - By^2 + \mathcal{O}(y^4).$$

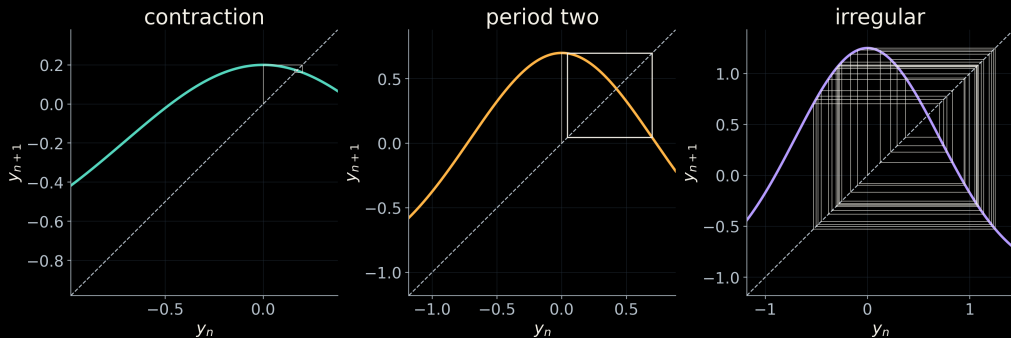
The critical orbit separates three regimes



$$(A, B) = (-0.8, 1.0), (-1, 1.7), (-1, 2.25)$$

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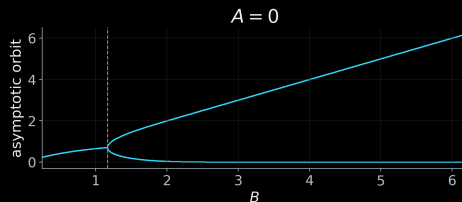
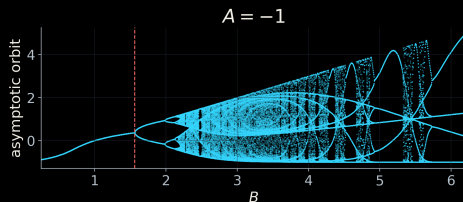
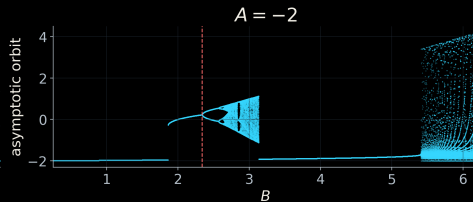
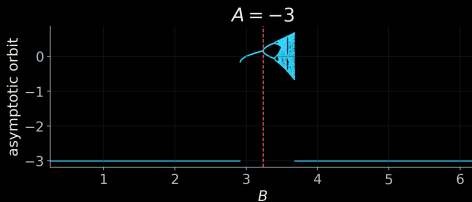
Cobwebs explain the same transitions



Diagonal = equilibria; vertical–horizontal steps = iteration.

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Thresholds anchor numerical slices



coral = analytic first doubling • **NUMERICAL DIAGNOSTIC**

Gaussian maps are S -unimodal

Structural theorem

For $B > 0$, the map $T(y) = A + Be^{-y^2}$ is C^∞ , unimodal, and has a unique nondegenerate quadratic critical point $c = 0$. Moreover,

$$S(T)(y) := \frac{T'''(y)}{T'(y)} - \frac{3}{2} \left(\frac{T''(y)}{T'(y)} \right)^2 = -2y^2 - \frac{3}{2y^2} < 0, \quad y \neq 0.$$

quadratic fold + negative Schwarzian $\implies$ classical
unimodal theory

The Schwarzian calculation

$$\begin{aligned}T'(y) &= -2Bye^{-y^2}, & T''(y) &= 2B(2y^2 - 1)e^{-y^2}, \\T'''(y) &= 4By(3 - 2y^2)e^{-y^2}.\end{aligned}$$

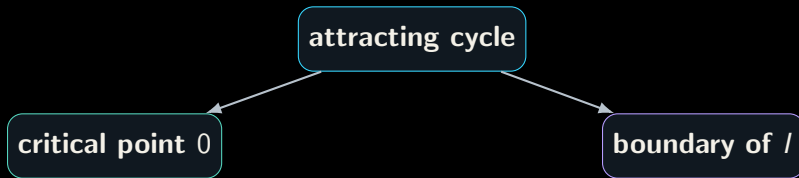
For $y \neq 0$,

$$S(T) = \frac{T'''}{T'} - \frac{3}{2} \left(\frac{T''}{T'} \right)^2 = -2y^2 - \frac{3}{2y^2} < 0.$$

Singer reduces the search

Singer consequence

For a C^3 unimodal interval map with negative Schwarzian derivative, the immediate basin of every attracting periodic orbit contains either a critical point or a boundary point of the invariant interval.



If the basin avoids ∂I , it contains 0: at most one such attractor exists.

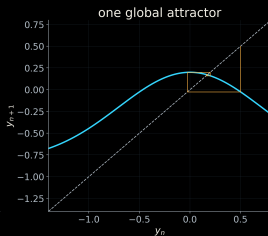
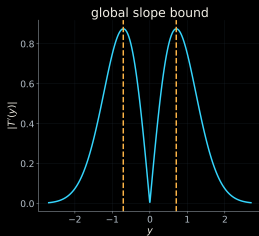
Boundary basins matter

Singer's theorem does *not* say that every attractor must attract the critical point: a boundary point may lie in its immediate basin.

Family-specific conclusion

If an attracting periodic orbit has immediate basin disjoint from ∂I , then that basin contains 0. Consequently, two such attracting cycles cannot coexist.

A threshold guarantees simplicity



Global contraction

$$\sup_y |T'(y)| = B\sqrt{\frac{2}{e}}.$$

If $0 < B < \sqrt{e/2}$, Banach's theorem gives a unique fixed point attracting every real orbit.

Fixed-point bifurcations are closed curves

Bifurcation loci

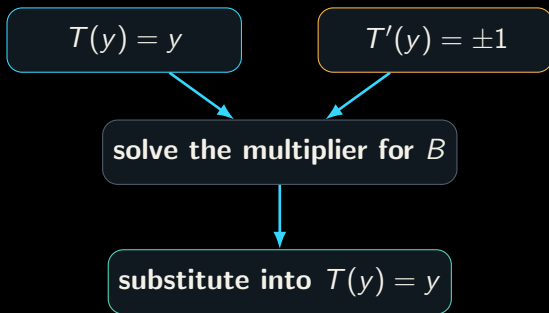
Using the fixed point y as parameter,

saddle-node: $A_{\text{sn}}(y) = y + \frac{1}{2y}, \quad B_{\text{sn}}(y) = -\frac{e^{y^2}}{2y}, \quad y < 0,$

period-doubling: $A_{\text{pd}}(y) = y - \frac{1}{2y}, \quad B_{\text{pd}}(y) = \frac{e^{y^2}}{2y}, \quad y > 0.$

The local bifurcation parameter is the fixed point itself.

Two equations generate both loci



$$B = \mp \frac{e^{y^2}}{2y}, \quad A = y - Be^{-y^2}.$$

The threshold is the cusp height

Along the saddle-node curve,

$$B_{\text{sn}}(y) = -\frac{e^{y^2}}{2y}, \quad y < 0.$$

Differentiation gives the unique minimum at $y = -1/\sqrt{2}$, hence

$$\min_{y < 0} B_{\text{sn}}(y) = \sqrt{\frac{e}{2}}, \quad A_{\text{sn}}(-1/\sqrt{2}) = -\sqrt{2}.$$

Thus the loss of global contraction occurs exactly at the saddle-node cusp.

Fixed- A thresholds

$$y_{\text{pd}}(A) = \frac{A + \sqrt{A^2 + 2}}{2},$$

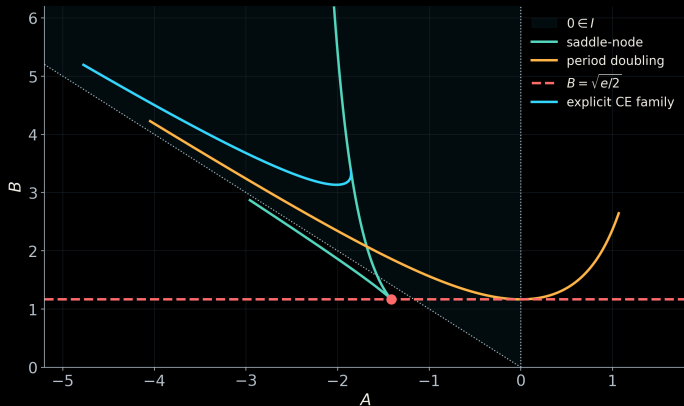
$$B_{\text{pd}}(A) = \frac{e^{y_{\text{pd}}^2}}{2y_{\text{pd}}},$$

$$y_{\text{sn}}^{\pm}(A) = \frac{A \pm \sqrt{A^2 - 2}}{2},$$

$$B_{\text{sn}}^{\pm}(A) = -\frac{e^{(y_{\text{sn}}^{\pm})^2}}{2y_{\text{sn}}^{\pm}}.$$

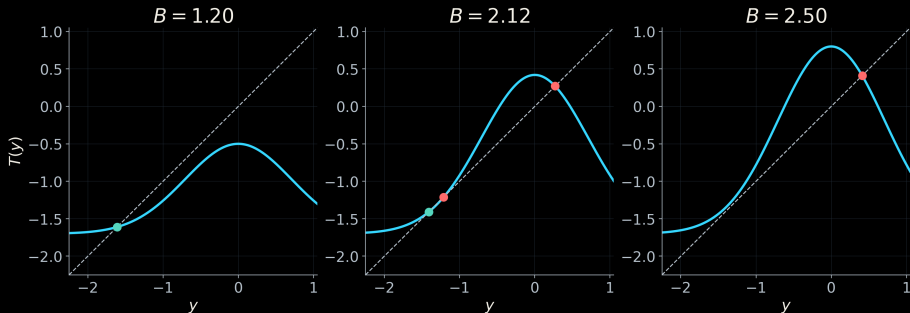
- ▶ one period-doubling threshold for every $A \in \mathbb{R}$;
- ▶ two saddle-node thresholds for $A < -\sqrt{2}$, merging at the cusp.

Exact curves organize parameter space



- ▶ cusp at $(-\sqrt{2}, \sqrt{e/2})$
- ▶ cusp height equals the contraction threshold
- ▶ $-B \leq A \leq 0$ places 0 in I
- ▶ cyan curve: exact CE family

Only one or three fixed points can occur



Fixed-point count

For $B > 0$, the equation $T(y) = y$ has at most three real solutions. Away from the saddle-node locus, exactly one or three simple fixed points occur.

At most three fixed points

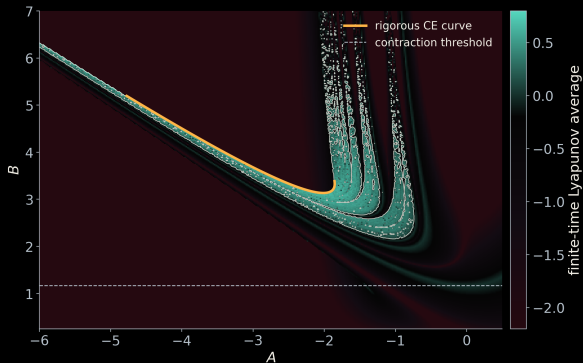
Let $F(y) = T(y) - y$. Then

$$F'(y) = -2Bye^{-y^2} - 1.$$

On $(-\infty, 0)$, the function $2B(-y)e^{-y^2}$ is unimodal, so F' has at most two zeros.
On $(0, \infty)$, $F'(y) < 0$.

F has at most two turning points, hence at most three zeros.

The CE curve enters an expanding region

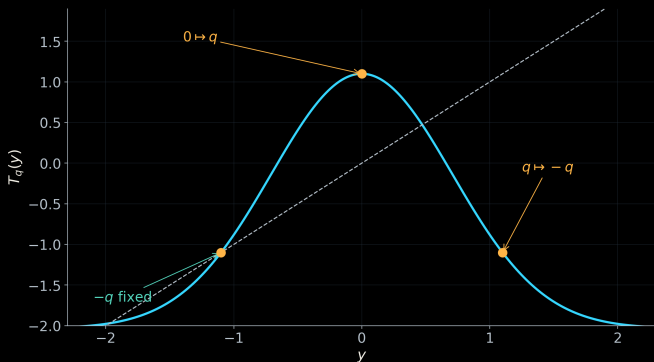


- ▶ color: finite-time Lyapunov average
- ▶ gold: rigorous CE curve
- ▶ white: zero-growth contour

INTERPRETATION

Finite-time colors diagnose;
the gold curve is exact.

An itinerary yields exact expansion



$$0 \mapsto q \mapsto -q \mapsto -q$$

$$A(q) = -\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}},$$

$$B(q) = \frac{2q}{1 - e^{-q^2}}.$$

Deriving the CE curve

Impose

$$T_q(0) = q, \quad T_q(q) = -q.$$

Equivalently,

$$A + B = q, \quad A + Be^{-q^2} = -q.$$

Solving gives

$$A(q) = -\frac{q(1 + e^{-q^2})}{1 - e^{-q^2}}, \quad B(q) = \frac{2q}{1 - e^{-q^2}}.$$

The same equations imply $T_q(-q) = -q$.

The expansion threshold is explicit

Explicit Collet–Eckmann family

Let $q_* > 0$ be the positive solution of

$$e^{q_*^2} = 1 + 4q_*^2, \quad q_* \approx 1.52861473.$$

For $0 < q < q_*$, define $T_q(y) = A(q) + B(q)e^{-y^2}$. With $y_0 = T_q(0) = q$,

$$|(T_q^n)'(y_0)| = \Lambda(q)^n, \quad \Lambda(q) = \frac{4q^2}{e^{q^2} - 1} > 1.$$

A closed form for q_*

Writing $t = q_*^2$, the equation $e^t = 1 + 4t$ can be rearranged into Lambert- W form.

$$q_* = \sqrt{-W_{-1}\left(-\frac{e^{-1/4}}{4}\right) - \frac{1}{4}} \approx 1.52861473$$

The branch W_{-1} gives the nonzero positive solution.

Why the itinerary expands

Along the explicit curve,

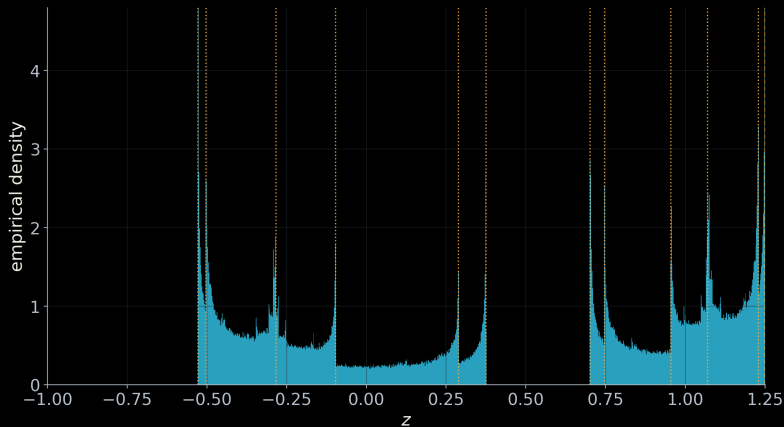
$$|T'_q(q)| = |T'_q(-q)| = 2B(q)qe^{-q^2} = \frac{4q^2}{e^{q^2} - 1} = \Lambda(q).$$

Since $q \mapsto -q \mapsto -q$,

$$|(T_q^n)'(q)| = \Lambda(q)^n.$$

The inequality $\Lambda(q) > 1$ is exactly $e^{q^2} < 1 + 4q^2$, giving $0 < q < q_*$.

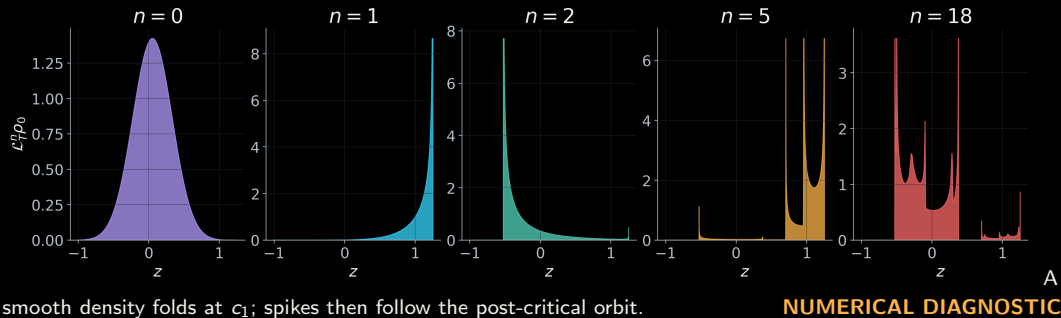
Density peaks align with the critical orbit



gold markers are $c_n = T^n(0)$.

$A = -1, B = 2.25$;
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Transfer iterates transport the spikes



Expansion produces a physical measure

Statistical consequence

Let $T : I \rightarrow I$ be the normalized Gaussian map with $0 \in I$. If the critical orbit is nonperiodic and

$$|(T^n)'(T(0))| \geq C\Lambda^n, \quad C > 0, \quad \Lambda > 1,$$

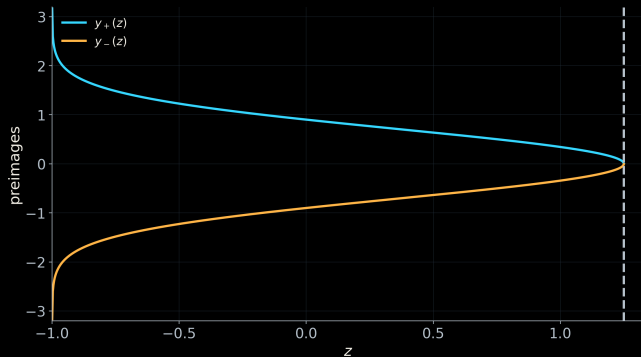
then T admits an ergodic, absolutely continuous invariant probability measure

$$d\mu = \rho(y) dy, \quad \rho \in L^1(I), \quad \rho \geq 0.$$

Its basin has positive Lebesgue measure.

The Gaussian itinerary proves CE; classical one-dimensional theory supplies the acip.

Inverse branches make transport explicit



For $A < z < A + B$,

$$y_{\pm}(z) = \pm \sqrt{-\log\left(\frac{z - A}{B}\right)}.$$

Both branches merge at the
critical value

$$c_1 = A + B.$$

The transfer operator is closed form

Perron–Frobenius formula

For $A < z < A + B$ and every integrable density ρ ,

$$(\mathcal{L}_T \rho)(z) = \frac{\rho(y_+(z)) + \rho(y_-(z))}{2|z - A|\sqrt{-\log((z - A)/B)}}.$$

Outside $(A, A + B)$, $\mathcal{L}_T \rho = 0$ almost everywhere.

Invariant density: $\rho = \mathcal{L}_T \rho.$

Deriving the transfer operator

At a preimage $y_{\pm}(z)$,

$$Be^{-y_{\pm}(z)^2} = z - A.$$

Hence

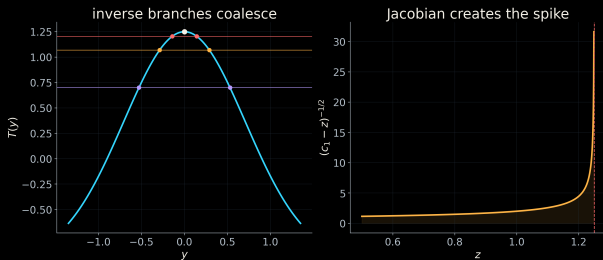
$$|T'(y_{\pm}(z))| = 2|z - A| |y_{\pm}(z)| = 2|z - A| \sqrt{-\log((z - A)/B)}.$$

Substitution into

$$(\mathcal{L}_T \rho)(z) = \sum_{T(y)=z} \frac{\rho(y)}{|T'(y)|}$$

gives the formula in the main talk.

The first spike obeys a universal law



First singularity

If an invariant density ρ is finite and continuous at 0, with $\rho(0) > 0$, then

$$\rho(z) \sim \frac{\rho(0)}{\sqrt{B}} \frac{1}{\sqrt{A+B-z}}$$

as $z \rightarrow (A+B)^-$.

Deriving the first spike

Write $\varepsilon = A + B - z \downarrow 0$. Then

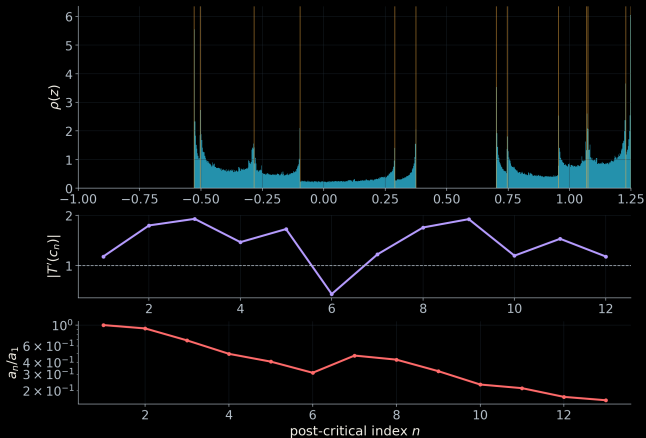
$$u(z) = \sqrt{-\log\left(1 - \frac{\varepsilon}{B}\right)} \sim \sqrt{\frac{\varepsilon}{B}}.$$

Continuity at the critical point gives

$$\rho(u) + \rho(-u) \rightarrow 2\rho(0), \quad 2(z - A)u \sim 2\sqrt{B\varepsilon}.$$

Therefore $\rho(z) \sim \rho(0)/\sqrt{B(A + B - z)}$.

Expansion controls spike amplitudes



Propagation law

Under the bounded-remainder and noncriticality hypotheses,

$$a_{n+1} = \frac{a_n}{\sqrt{|T'(c_n)|}},$$

$$a_n = \frac{\rho(0)}{\sqrt{B}} |(T^{n-1})'(c_1)|^{-1/2}.$$

Finite spike decomposition

If $c_1, \dots, c_N$ are distinct and the stated regularity hypotheses hold, then locally

$$\rho(z) = \psi_N(z) + \sum_{n=1}^N a_n K_n(z),$$

where ψ_N is bounded and

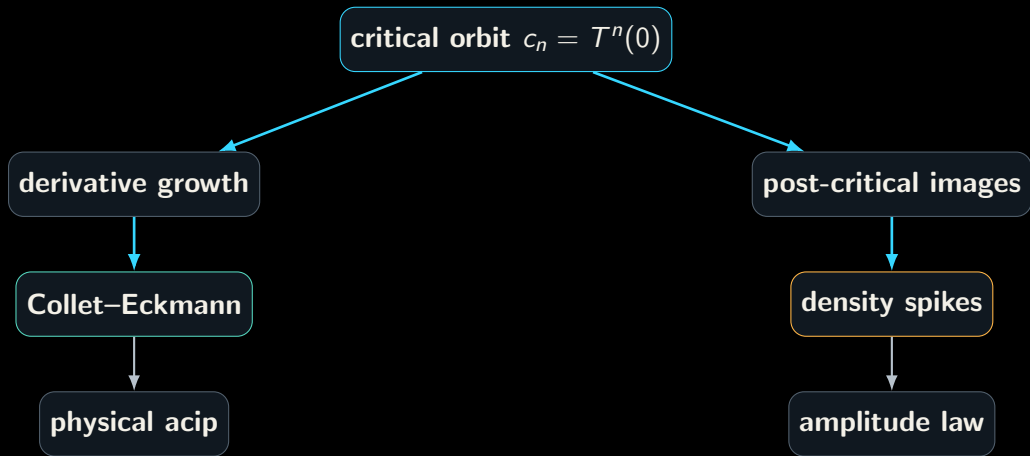
$$K_n(z) = \begin{cases} [\sigma_n(z - c_n)]^{-1/2}, & \sigma_n(z - c_n) > 0, \\ 0, & \sigma_n(z - c_n) < 0. \end{cases}$$

The orientations satisfy $\sigma_1 = -1$ and $\sigma_{n+1} = \sigma_n \operatorname{sgn} T'(c_n)$.

Theorem context and scope

Claim	Status	Scope
Negative Schwarzian	direct computation	$y \neq 0$
Attractor restriction	Singer's theorem	critical or boundary basin
Explicit growth	exact preperiodic orbit	$0 < q < q_*$
Acip existence	classical CE theory	nonperiodic CE critical orbit
Density spikes	local transfer analysis	continuity and bounded remainder
Parameter landscapes	finite computation	diagnostic, not certification

One orbit links dynamics and statistics



Three explicit structures drive the theory

1. **Geometry is explicit.** Affine conjugacy produces a bounded quadratic fold with negative Schwarzian derivative.
2. **Bifurcations are analytic.** Contraction, saddle-node, period doubling, and an exact CE curve organize parameter space.
3. **Statistics retain the critical orbit.** The closed transfer operator generates square-root spikes along $\{T^n(0)\}$.

Quadratic universality meets Gaussian tractability.

Questions?



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